

august 2025



[noForth website](#)

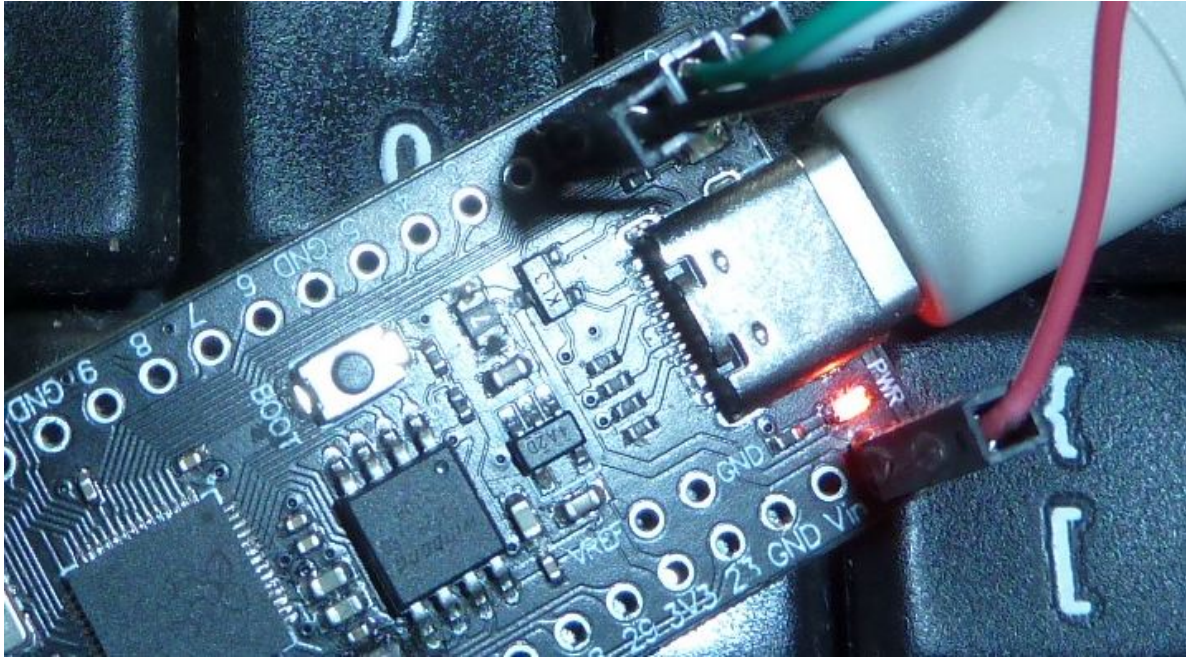
Pico RP2040 dev boards with noForth t(v)(#) (solo/duo)

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In this text we refer to these two documents:

- [RP2040 datasheet](#)
- [Armv6-M Architecture Reference Manual](#)

1. RP2040 Pico dev boards



Core : M0+ core with ARMv6-M ISA
Kit Contents: Various Pico Dev Boards

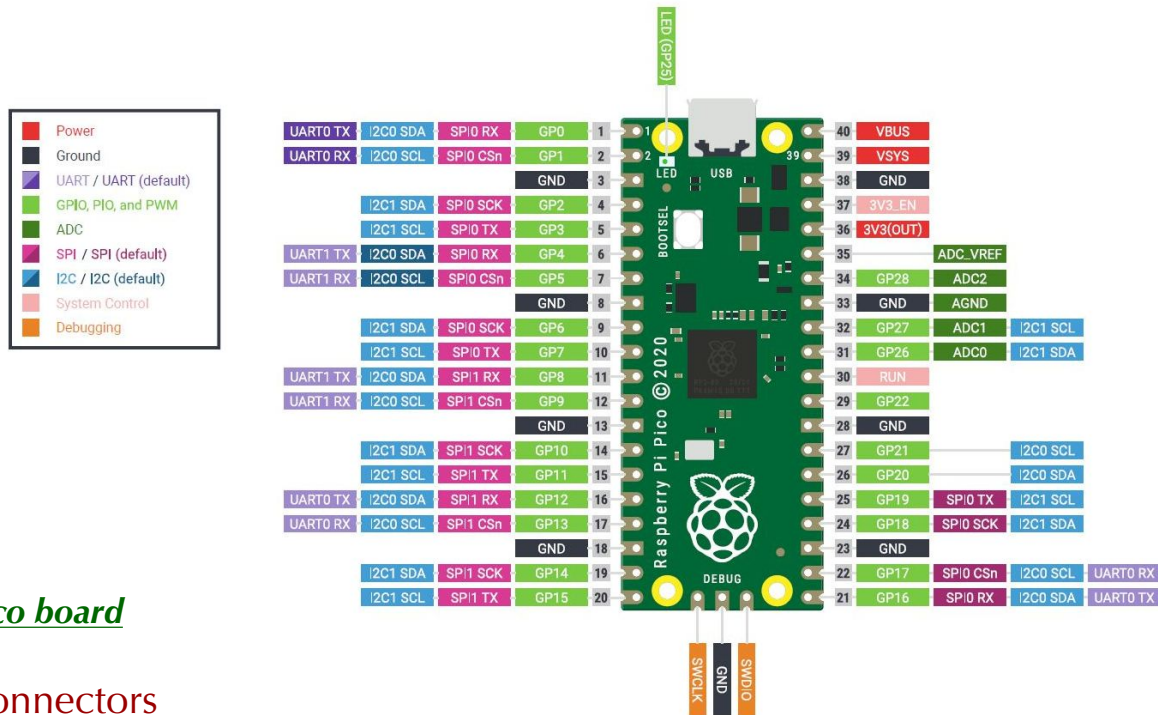
- Antratek: Pico dev board
- Antratek: Pico 2 dev board
- Kiwi: Pico dev board
- Kiwi: Pico 2 dev board
- YP-RP2040 board
- YP-RP2040 alternative board

i/o port connections pico

Between parens are the pin usage differences between the original Pico and the YD-RP2040

GPI0

GPI00	- UART0 TX	GPI016	- ...
GPI01	- UART0 RX	GPI017	- ...
GPI02	- ...	GPI018	- ...
GPI03	- ...	GPI029	- ...
GPI04	- UART1 TX	GPI020	- ...
GPI05	- UART1 RX	GPI021	- ...
GPI06	- ...	GPI022	- ...
GPI07	- ...	GPI023	- SMPS (WS2812)
GPI08	- ...	GPI024	- VBUS sense (switch)
GPI09	- ...	GPI025	- LED
GPI010	- ...	GPI026	- ...
GPI011	- ...	GPI027	- ...
GPI012	- ...	GPI028	- ...
GPI013	- ...	GPI029	- VSYS (...)
GPI014	- ...		
GPI015	- ...		



Pico board

Connectors

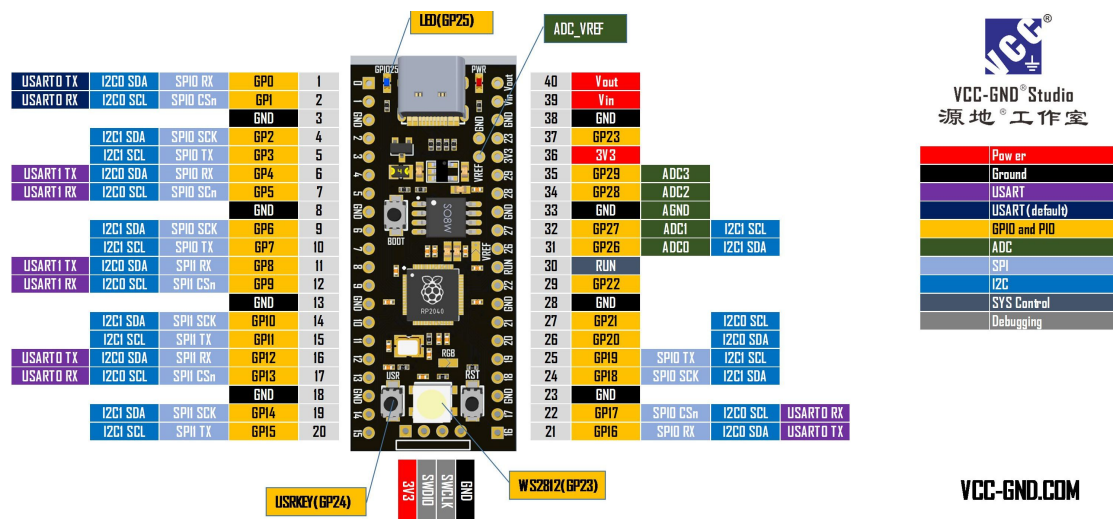
- J0 = i/o GPIO and GND
- J1 = i/o GPIO, RUN, GND, 5V, 3V3, 3V3_EN and ADC_VREF
- J2 = USB-C Programming connector
- J3 = Debug (SWD) port

Hardware Pico

- LED on GPIO25
- Switch BOOTSEL on QSPI_SS
- SPI Flash (16 to 128 Mbit)
- 12MHz crystal

Hardware YD-RP2040

- LED on GPIO25, power LED & WS2812 LED on GPIO23
- Three switches BOOTSEL on QSPI_SS, reset & GPIO24 (user)
- SPI Flash (32 to 128 Mbit)
- 12MHz crystal

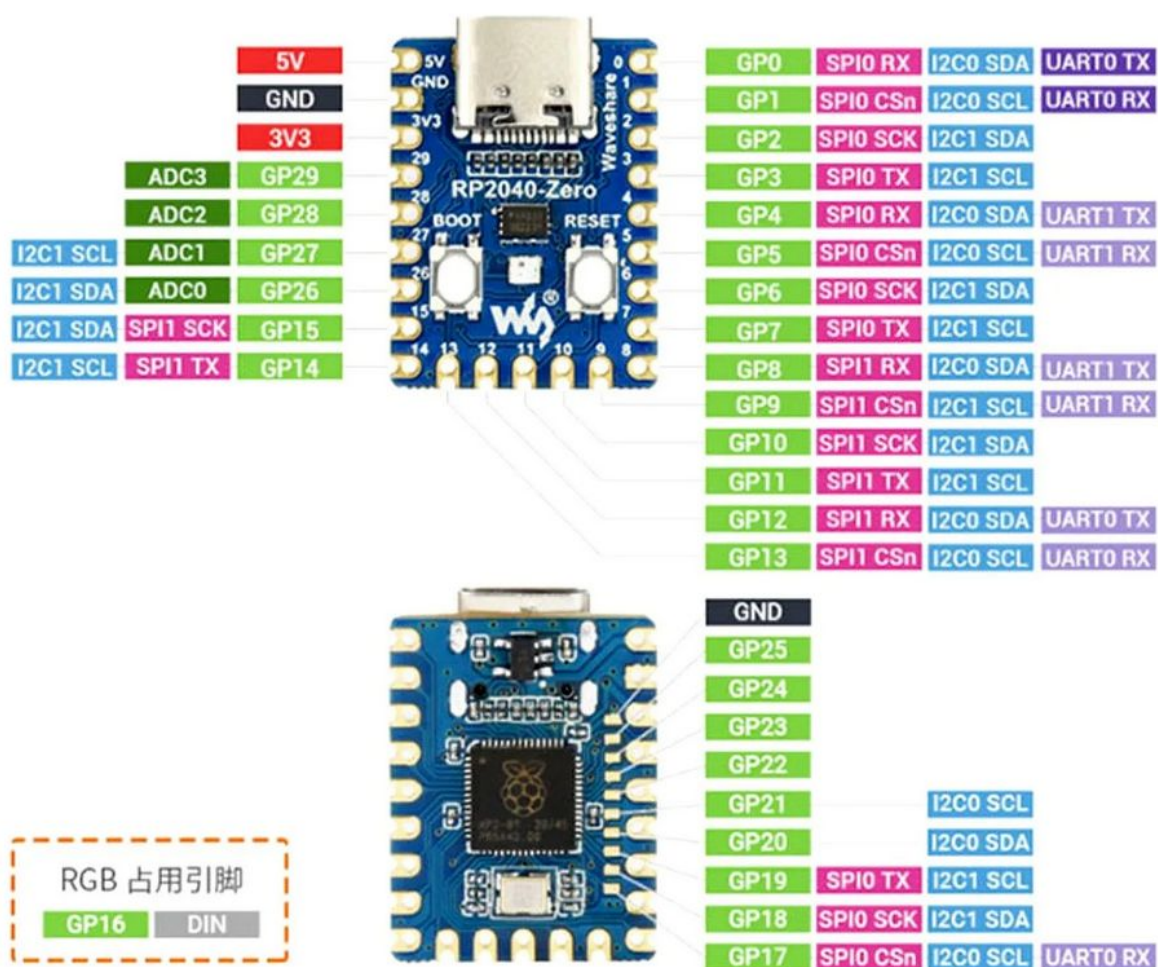


YD-RP2040 board

Only the board pins 35 & 37 are connected differently, see pictures.
pin 35 = GPIO29, pin 37 = GPIO23

Hardware RP2040-zero

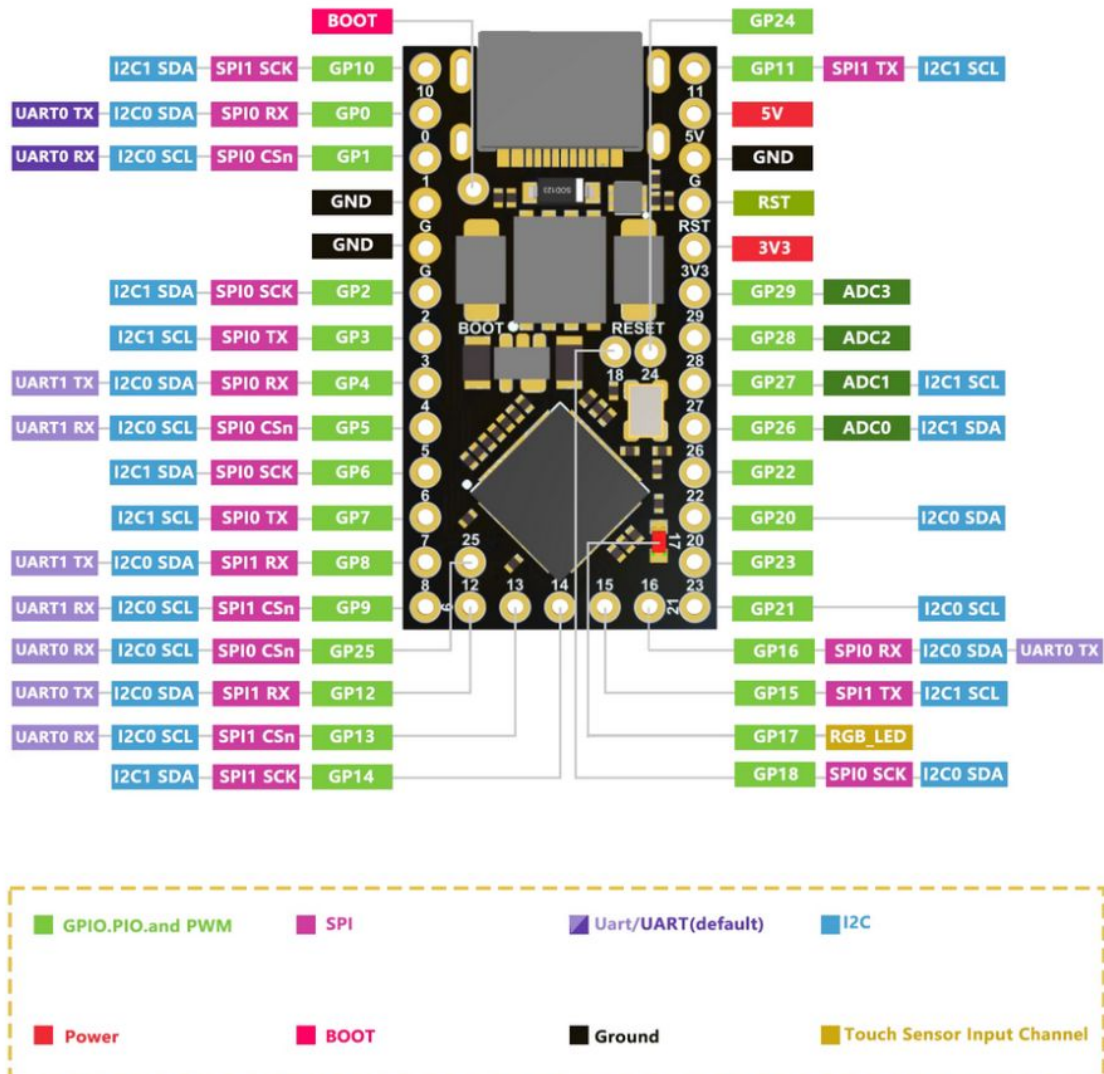
- WS2812 LED on GPIO16
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (16 Mbit)
- 12MHz crystal



RP2040-zero board

Hardware RP2040 ProMicro

- LED on GPIO17
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (16 to 128 Mbit)
- 12MHz crystal



RP2040 ProMicro board

2. I/O ports

I/O port addresses

The port registers are memory mapped.
First the base addresses for the I/O-port:

SIO base = D0000000
User bank = 40014000 (Two cells for each pin)

Port register offsets for SIO base

More functionality can be found from page 42ff.

Label	SIO base	Function
CPUID	00	Processor core ID
GPIO_IN	04	Input value for GPIO
GPIO_HI_IN	08	Input value for QSPI
GPIO_OUT	10	Output value
GPIO_SET	14	Output value set
GPIO_CLR	18	Output value clear
GPIO_XOR	1C	Output value toggle
Etc.		

Values for port control IO_BANK0_BASE registers

On page 13 & 14 of the RP2040 datasheet, you find a complete list of all functions that can be chosen for each GPIO pin. The pin control functions starts at offset+4 for GPIO0 and are increased by 8 for each next pin.

GPIO	Function								
	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PI00	PI01		USB OVCUR DET
1	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PI00	PI01		USB VBUS DET
2	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PI00	PI01		USB VBUS EN
3	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PI00	PI01		USB OVCUR DET
4	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PI00	PI01		USB VBUS DET

- 1: SPI-0 or 1
- 2: UART-0 or 1
- 3: I2C-0 or 1
- 4: PWM-0 to PWM-7
- 5: Standard I/O (Reset state)
- 6: Programmable I/O (PI0-0)
- 7: Programmable I/O (PI0-1)
- 8: Clock in or out
- 9: USB-bus

The reset value for all port control registers is 1F. More info in the 'RP2040 datasheet.PDF' from page 236ff.

3. RS232/USB driver

The USB chip for the SEED board is a dongle with the PL2303TA Prolific USB-chip. This chip needs a specific driver under Windows XP/7/8/10/11. Execute this file: [PL2303-Prolific_DriverInstaller_v1200.exe](#). The default baudrate for the RP2040 controller with noForth is 115200 baud.

There is also a USB CDC driver available that runs on top of [noForth tv#](#). This is a retrofittable option. An off-the-shelf UF2 is also available for simplicity.

4a. noForth t solo (multitask) memory map

RAM 21000000 - 21040000

```
21000000 IVECS      \ 48 vectors
      ... IVECS/    \ Vector table end
210000D0 HOT        \ Uhere starts here
21000200 ORIGIN     \ HERE starts here
\ ... system ...
2100xxxx HERE      \ Current systems end
2101F800 ...        \ Systems end
2101F800 FLYBUF     \ Flyer buffer (400 bytes)
2101FC00 FLYBUF/    \ End of flyer buffer
2101FD00 SP0        \ Data stack (100 bytes grows down)
2101FF00 RP0        \ Return stack (200 bytes grows down)
2101FF00 TIB        \ Input buffer (100 bytes)
21020000 TIB/
21040000 MEMTOP     \ End of RAM memory
```

RAM 20040000 - 20042000

8 kByte of free RAM memory in two 4 kByte banks.
The multitasker version uses a part of this memory for new tasks.

4b. noForth t duo (multitask) memory map

noForth t memory map core-0

RAM 21000000 - 21020000

```
21000000 IVECS      \ 48 vectors
      ... IVECS/    \ Vector table end
210000D0 HOT        \ Uhere starts here
21000200 ORIGIN     \ HERE starts here
\ ... system ...
2100xxxx HERE      \ Current systems end
2101F800 ...        \ Systems end
2101F800 FLYBUF     \ Flyer buffer (400 bytes)
2101FC00 FLYBUF/    \ End of flyer buffer
2101FD00 SP0        \ Data stack (100 bytes grows down)
2101FF00 RP0        \ Return stack (200 bytes grows down)
2101FF00 TIB        \ Input buffer (100 bytes)
21020000 TIB/
21020000 MEMTOP     \ End of core-0 RAM memory
```

noForth t memory map core-1

RAM 21020000 - 21040000

```
21020000 IVECS      \ 48 vectors
      ... IVECS/    \ Vector table end
210200D0 HOT        \ Uhere starts here
21020200 ORIGIN     \ HERE starts here
\ ... system ...
2102xxxx HERE      \ Current systems end
2103F800 ...        \ Systems end
2103F800 FLYBUF     \ Flyer buffer (400 bytes)
2103FC00 FLYBUF/    \ End of flyer buffer
2103FD00 SP0        \ Data stack (100 bytes grows down)
2103FF00 RP0        \ Return stack (200 bytes grows down)
2103FF00 TIB        \ Input buffer (100 bytes)
21040000 TIB/
21040000 MEMTOP     \ End of core-1 RAM memory
```

RAM 20040000 - 20042000

8 kByte of free RAM memory in two 4 kByte banks.

The multitasker version uses a part of this memory for new tasks.

4c. FLASH ROM (XIP mode) 10000000 - 1003FFFF

```
10000000 Secondary boot code, loads image-1 on startup
10000100 Image-1 \ Cold start image invoked by COLD to
10043000 Image-2 \ Spare image invoked by COLD2
10083000 Free    \ Start of free flash space
```

- TIB/ is a changeable uvalue (Udata).
- Udata is saved by FREEZE too.
- At start-up (and reset or COLD) noForth moves from Flash to RAM.

5. Interrupt vector table

See 'RP2040 datasheet.PDF' for more details (page 60ff) and 'ARMv6-M Architecture Reference Manual.pdf' from page 218ff. The table is pointed to by the VTOR register at PPB_BASE + ED08. Note that the low 8 bits of VTOR are always zero, so it's 256 bytes page aligned.

RP2040 exceptions:

0000	00	- Length of noForth image
0004	01	- Reset vector
0008	02	- NMI handler
000C	03	- Hard fault handler
0010	04	- MPU fault handler
0014	05	- Reserved
0018	06	- Reserved
001C	07	- Reserved
0020	08	- Reserved
0024	09	- Reserved
0028	10	- Reserved
002C	11	- Supervisor call handler
0030	12	- Reserved
0034	13	- Reserved
0038	14	- Pend supervisor handler
003C	15	- SysTick handler

NVIC IRQ interrupts:

0040	00	- Timer IRQ 0
0044	01	- Timer IRQ 1
0048	02	- Timer IRQ 2
004C	03	- Timer IRQ 3
0050	04	- PWM IRQ wrap
0054	05	- USB vontrol IRQ
0058	06	- XIP IRQ
005C	07	- PIO 0 IRQ 0
0060	08	- PIO 0 IRQ 1
0064	09	- PIO 1 IRQ 0
0068	10	- PIO 1 IRQ 1
006C	11	- DMA IRQ 0
0070	12	- DMA IRQ 1
0074	13	- IO IRQ bank 0
0078	14	- IO IRQ QSPI
007C	15	- SIO IRQ proc 0
0080	16	- SIO IRQ proc 1
0084	17	- Clock IRQ
0088	18	- SPI 0 IRQ
008C	19	- SPI 1 IRQ
0090	20	- UART 0 IRQ
0094	21	- UART 1 IRQ
0098	22	- ADC IRQ fifo
009C	23	- I2C 0 IRQ
00A0	24	- I2C 1 IRQ
00A4	25	- RTC IRQ
00A8	26	- Connected to zero
00AC	27	- Connected to zero
00B0	28	- Connected to zero
00B4	29	- Connected to zero
00B8	30	- Connected to zero
00BC	31	- Connected to zero